

ANALYSTLAB AFRICA EXPERIENCE LAB**HEALTHCONNECT WEEK 6****1. Purpose of This Document**

This document summarises key findings from Week 6 HealthConnect Data Analytics work and provides them to the Data Science track as input for model improvement, feature evaluation, error analysis and validation.

The objective of this cross-track integration is to determine whether the patterns identified through advanced data analysis can provide useful support for:

- Feature selection and refinement
- Feature engineering and interaction testing
- Model interpretation and validation
- Error analysis
- Identification of high-risk patient and appointment segments
- Comparison of model predictions with observed analytical patterns

The broader HealthConnect project objective is to explore how data and AI can help reduce missed appointments and improve patient support.

2. Week 5 to Week 6 Analytical Progression

During Week 5, the Data Analytics work focused on:

- Data preparation and exploratory analysis
- KPI development
- Analysis of appointment outcomes and no-show behaviour
- Identification of factors associated with missed appointments
- Development of an initial Power BI dashboard
- Business insights and recommendations

During Week 6, the work progressed from general exploratory analysis toward deeper investigation, KPI validation and decision support. The focus was placed on identifying high-impact patterns and analytical findings that could also be relevant to predictive modelling.

3. Key Analytical Findings Relevant to Data Science**Finding 1: Previous No-Show History Is Strongly Associated With Future No-Shows**

Previous no-show history emerged as one of the strongest observed indicators associated with appointment attendance behaviour. The no-show rate increased as the number of previous no-shows increased.

Previous No-Shows	Approximate No-Show Rate
0	43.5%

1	53.5%
2	59.4%
3+	68.8%

Analytical interpretation:

Patients with a history of missed appointments showed progressively higher observed no-show rates. This suggests that previous no-show history may be a valuable variable to evaluate in predictive modelling.

Potential Data Science contribution:

- Evaluate previous_no_shows as a candidate predictive feature.
- Assess its relative importance and contribution to model performance.
- Test whether the relationship with the target is linear or non-linear.
- Investigate interactions with booking lead time, reminder status and distance to clinic.

Finding 2: Longer Booking Lead Times Are Associated With Higher No-Show Rates

Appointments booked further in advance showed higher no-show rates. Appointments booked approximately 31–60 days in advance recorded a no-show rate of approximately 60.5%, making this one of the highest-risk booking lead-time segments.

Analytical interpretation:

Longer periods between booking and appointment dates were associated with a greater likelihood of missed appointments. This is an observed relationship and should not automatically be interpreted as causal.

Potential Data Science contribution:

- Evaluate booking_lead_days as a predictive feature.
- Compare the continuous lead-time variable with grouped lead-time bands.
- Test whether specific lead-time ranges are particularly informative for prediction.
- Investigate interactions between booking lead time and previous no-shows or reminders.

Finding 3: Reminder Status Is Associated With Lower Observed No-Show Rates

Appointments with reminders showed a lower observed no-show rate than appointments without reminders.

Reminder Status	Approximate No-Show Rate
No Reminder	51.4%
Reminder Sent	47.4%

The observed difference is approximately 4 percentage points, equivalent to an approximate relative reduction of 7.8%.

Analytical interpretation:

The relationship should be interpreted carefully because the data is observational. The analysis does not establish that reminders alone caused the difference in no-show rates.

Potential Data Science contribution:

- Include reminder_sent as a candidate feature for evaluation.
- Assess whether it improves model performance after accounting for other variables.
- Investigate whether reminder patterns differ across high-risk patient groups.
- Explore interactions between reminders and booking lead time, distance or previous no-show history.

Finding 4: Distance to the Clinic May Be Associated With No-Show Behaviour

The analysis identified variation in no-show rates across distance bands. Patients travelling approximately 20 km or more recorded an observed no-show rate of approximately 58%.

Analytical interpretation:

Travel distance may be associated with barriers affecting appointment attendance. However, the available data does not establish the underlying cause of this relationship.

Potential Data Science contribution:

- Evaluate distance_to_clinic_km as a candidate predictive feature.
- Compare use of the continuous distance variable with grouped distance bands.
- Assess whether distance provides meaningful predictive value.
- Investigate possible interactions between distance and reminder status.

Finding 5: Previous No-Show History Identifies a High-Risk Segment

Appointments associated with patients who had at least one previous no-show recorded an overall no-show rate of approximately 53.5%, compared with approximately 44% among those with no previous no-show history.

Business significance:

- Targeted appointment reminders
- Appointment confirmation or follow-up communication
- High-risk patient flagging
- Additional attendance support for patients identified as higher risk

4. Additional Patterns Identified

The Week 6 analysis also investigated no-show behaviour across the following dimensions:

- Appointment type
- Reminder channel
- Age group
- Appointment time
- Appointment day

These variables may provide additional modelling value and should be evaluated according to their contribution to predictive performance rather than descriptive differences alone.

Important note: Not every variable that shows a difference in descriptive analysis will necessarily improve predictive performance. Model evaluation should therefore consider feature importance, predictive contribution, redundancy and possible interactions.

5. Suggested Data Science Questions

1. Which of the analytical findings correspond to the strongest predictive features?
2. Does previous_no_shows significantly improve model performance?
3. Does categorising booking_lead_days improve prediction compared with using the continuous variable?
4. Does distance_to_clinic_km provide meaningful predictive value?
5. Does reminder_sent improve model performance after accounting for other variables?
6. Are there important interactions between previous no-shows and booking lead time, reminders or distance?
7. Do the model's highest-risk predictions align with the high-risk segments identified during the Data Analytics work?
8. Are there important analytical patterns that the model does not capture effectively?

6. Requested Data Science Feedback

To support meaningful Week 6 cross-track integration, feedback is requested on the following areas:

A. Findings Used

Which analytical findings or variables were considered during model improvement?

B. Modelling Impact

Did any of the findings influence:

- Feature selection
- Feature engineering or refinement
- Model interpretation
- Error analysis
- Model comparison or validation

C. Model Validation

Did the model identify similar high-risk patterns to those observed during the Data Analytics analysis?

D. Additional Analytical Requirements

Are there additional analyses, segment investigations or validation checks that would be useful for the modelling process?

7. Cross-Track Integration Objective

Data Analytics Contribution	Expected Data Science Contribution
Validated KPIs; high-risk patient and appointment segments; evidence of relationships associated with no-show behaviour; candidate feature considerations; business context for interpreting model outputs.	Feedback on the relevance of findings; variables or relationships evaluated; whether findings influenced modelling decisions; alignment between model outputs and observed analytical patterns.

8. Expected Integration Outcome

This integration is intended to connect descriptive and diagnostic analysis with predictive modelling. The Data Analytics findings provide evidence and business context that may support:

- Identification of potentially relevant model features
- Understanding of high-risk patient and appointment segments
- Interpretation of model predictions
- Evaluation of whether model outputs align with observed data patterns

The Data Science work can, in turn, help determine whether these observed patterns can be used effectively for appointment no-show prediction and whether the model captures the key risks identified through analysis.

9. Data and Analytical Limitations

- The dataset represents a simulated HealthConnect environment and may not fully reflect real patient behaviour.
- Observed relationships do not establish causation.
- Reminder effectiveness cannot be confirmed as causal without controlled experimentation or additional causal analysis.
- Important real-world factors influencing attendance may not be represented in the dataset.
- The predictive usefulness of variables should be determined through model evaluation rather than descriptive patterns alone.
- High no-show rates within specific segments may be influenced by other related variables or confounding factors.

10. Proposed Next Steps

9. The Data Science intern reviews the analytical findings and identifies relevant variables or relationships for model improvement and validation.
10. The Data Science intern provides feedback on which findings were considered and whether any influenced modelling decisions.
11. Where appropriate, the Data Analytics and Data Science findings are compared to determine whether high-risk analytical segments align with model predictions.
12. Both tracks document the integration activity and resulting change as evidence for the Week 6 submission.
13. Any unresolved analytical or modelling questions are carried forward into Week 7 testing and refinement.

Power Bi Dashboard (Health Connect Dataset)

